Question: Given a function $f(x)$ that is continuously differentiable, can we find constants c_0, c_1, c_2, ... such that

$$f(x) = \sum_{n=0}^{\infty} c_n x^n \ ?$$

Power series takes advantage of things that look like what geometric series converge to, this is broader?

$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$

How do I determine this constant?

Plug in something for x that isolates this

↳ $f(0) = c_0 + (c_1(0)) + \dots$

↳ $f(0) = c_0$ → $0!\, c_0$

To find c_1, differentiate so you have it as a constant

↳ $f'(x) = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$

↳ $f'(0) = c_1$ → $1!\, c_1$

Then you can do this w/ c_2, ... since the derivative of a power series is another power series so you can keep doing this

↳ $f''(x) = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$

↳ $f''(0) = 2c_2$ → $2!\, c_2$

You can keep doing this...

$f'''(0) = 6c_3$ → $3!\, c_3$

$f''''(0) = 24c_4$

$f''''(0) = (4\cdot3\cdot2\cdot1)c_4$

$f''''(0) = 4!\, c_4$

So the question then becomes

$f^{(n)}(0) = \ ? \ c_n$

$f^{(n)}(0) = n!\, c_n \longrightarrow c_n = \dfrac{f^{(n)}(0)}{n!}$

There are functions where you cannot do this:

$f(x) = \sqrt{x}$

↳ $f(0) = 0$

$f'(x) = \frac{1}{2} \cdot \frac{1}{\sqrt{x}}$

↳ $f'(0)$ = Dividing by 0

So long as f(x) has derivatives that don't "blowup" at 0 (say, dividing by 0 at a given derivative) . . .

A **Taylor series about** $\underbrace{x = a}_{\text{centered at } a}$ is defined as

Derivative, n refers to 1st, 2nd, 3rd, etc...

so instead of evaluating the derivative at 0, evaluated at center

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

A **Maclaurin series** is a special case of a Taylor series with $a = 0$, i.e.,

centered at 0

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Example: Find a Maclaurin series (i.e., a Taylor series about $x = 0$) for $f(x) = \ln(1 + x)$.

$f^{(n)}(x)$	$f^{(n)}(0)$
$f(x) = \ln(1+x)$	$f(0) = \ln(1) = 0$
$f'(x) = (1+x)^{-1}$	$f'(0) = \frac{1}{1+0} = 1 = 1!$
$f''(x) = -1(1+x)^{-2}$	$f''(0) = -1 = -(1)!$
$f'''(x) = 2(1+x)^{-3}$	$f'''(0) = 2 = 2!$
$f''''(x) = -6(1+x)^{-4}$	$f''''(0) = -6 = -(3!)$
$f'''''(x) = 24(1+x)^{-5}$	$f'''''(0) = 24 = 4!$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \underbrace{\frac{f'(0)}{1!}}_{1! = 1} x + \underbrace{\frac{f''(0)}{2}}_{2! = 2} x^2 + \cdots$$

$$= 0 + 1x - \frac{x^2}{2} + \frac{2!}{3!}x^3 - \frac{3!}{4!}x^4 + \frac{4!}{5!}x^5 - \cdots$$

Simplifying

$$= 0 + x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots$$

$\frac{2!}{3!} = \frac{2!}{(3)2!} = \frac{1}{3}$

$$f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

can do this because first term is 0?

Making into a full power series based on intuition

Let $f(x) = e^x$, Find Mclaurin series

$f^{(n)}(x)$	$f^{(n)}(0)$
$f(x) = e^x$	$f(0) = 1$
$f'(x) = e^x$	$f'(0) = 1$
$f''(x) = e^x$	$f''(0) = 1$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

Find the radius of convergence

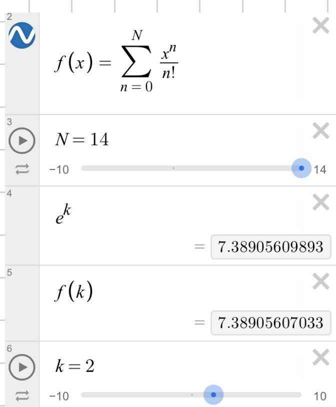
$$L = \lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right| = \lim_{n\to\infty} \left| \frac{1}{\underbrace{(n+1)!}_{=(n+1)n!}} \cdot \frac{n!}{1} \right| = \lim_{n\to\infty} \left|\frac{1}{n+1}\right| = 0$$

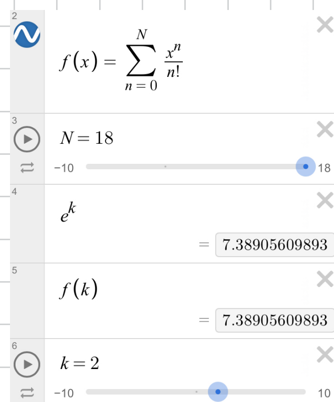
$$0 = L|x| < 1$$

↳ 0 × anything is always 0 which is always less than 1.

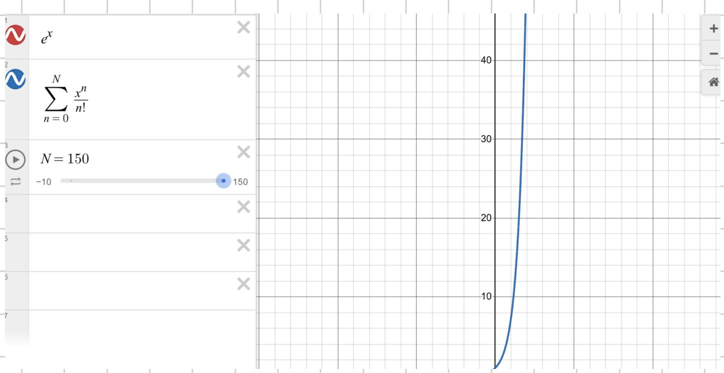
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for } (-\infty, \infty)$$

If K was 1, e^k and $f(x)$ would match up





the amount of terms you need for the partial sum varies based on x



Let $f(x) = \cos(x)$. Find Maclaurin Series (centered @ $x=0$)

Find ROC

$f^{(n)}(x)$	$f^{(n)}(0)$
$f(x) = \cos(x)$	$f(0) = 1$
$f'(x) = -\sin(x)$	$f'(0) = 0$
$f''(x) = -\cos(x)$	$f''(0) = -1$
$f'''(x) = \sin(x)$	$f'''(0) = 0$
$f''''(x) = \cos(x)$	$f''''(0) = 1$

odd ns give you 0

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \to \frac{1x^0}{0!} + \cancel{\frac{0x}{1!}} - \frac{1x^2}{2!} + \cancel{\frac{0x^3}{3!}} + \frac{x^4}{4!} + \ldots$$

$$\cos(x) = 1 \boxed{-} \frac{x^2}{2!} + \frac{x^4}{4!} \boxed{-} \frac{x^6}{6!} + \ldots$$

$$\cos(x) = \sum_{n=0}^{\infty} \underbrace{\boxed{(-1)^n}}_{\text{Alternating sign}} \frac{x^{2n}}{(2n)!}$$

only even terms

ROC of $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$

$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{(2(n+1))!} \cdot \frac{(2n)!}{(-1)^n} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(-1)\cancel{(-1)^n}}{(2n+2)(2n+1)\cancel{(2n)!}} \cdot \frac{\cancel{(2n)!}}{\cancel{(-1)^n}} \right|$$

$$= \lim_{n\to\infty} \left| \frac{(-1)}{(2n+2)(2n+1)} \right| = \lim_{n\to\infty} \frac{1}{(2n+2)(2n+1)} = 0$$

Since $L = 0$, ROC is infinite

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \text{ for } (-\infty, \infty)$$

the terms being even is why $\cos(-x) = \cos(x)$

Let $f(x) = \sin(x)$. Find Maclaurin Series (centered at $x=0$)

$f^{(n)}(x)$	$f^{(n)}(0)$
$f(x) = \sin(x)$	$f(0) = 0$
$f'(x) = \cos(x)$	$f'(0) = 1$
$f''(x) = -\sin(x)$	$f''(0) = 0$
$f'''(x) = -\cos(x)$	$f'''(0) = -1$
$f''''(x) = \sin(x)$	$f''''(0) = 0$

Even terms 0 out

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\text{Ln } \sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

Powers being odd is why $\sin(-x) = -\sin(x)$

ROC

$$L = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{(2n+3)!} \cdot \frac{(2n+1)!}{(-1)^n} \right| = \lim_{n\to\infty} \frac{\cancel{(2n+1)!}}{(2n+3)(2n+2)\cancel{(2n+1)!}}$$

$$= \lim_{n\to\infty} \frac{1}{(2n+3)(2n+2)} = 0$$

$L = 0$, so ...

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad \text{for } (-\infty, \infty)$$